Homework 13
Introduction to Complex Analysis
(MATH 4230-40), Spring 2022

Section 5.2 - Fractional Linear and Schwarz-Christoffel Transformations

1. Find fractional linear transformations f satisfying $f(z_i) = w_i$ for $i = 1, 2, 3$ if
   (a) $z_1 = -1$, $z_2 = 1$, $z_3 = 2$ and $w_1 = 0$, $w_2 = -1$, $w_3 = -3$
   (b) $z_1 = -1$, $z_2 = 1$, $z_3 = 2$ and $w_1 = -3$, $w_2 = -1$, $w_3 = 0$

2. Prove that any fractional linear transformation T with $c \neq 0$ can be written $T = T_4 \circ T_3 \circ T_2 \circ T_1$, where
$$T_1(z) = z + \frac{d}{c}, \quad T_2(z) = \frac{1}{z}, \quad T_3(z) = \frac{bc - ad}{c^2} z, \quad \text{and} \quad T_4(z) = z + \frac{a}{c}.$$